Examiner only

5. A biogas generator converts cow dung into biogas which can be burned to generate electricity.

(a) A farmer buys 3750 kWh of electricity per week from the National Grid.
He plans to buy a biogas generator which should reduce this to 750 kWh per week.

(i) Calculate how many units of electricity he would **save** per week. [1]

Units saved = ... kWh

(ii) Use the equation:

saving per week = units saved × cost per unit

to calculate the saving per week. One unit of electricity costs £0.20. [2]

Saving per week = £ ....

(iii) The biogas generator costs £150 000.

I. Use the equation:

$$\text{payback time} = \frac{\text{cost}}{\text{saving per week}}$$

to calculate the expected payback time in weeks. [1]

Payback time = ... weeks

II. Calculate the expected payback time in years. 1 year = 52 weeks. [1]

Payback time = ... years

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(b) The farmer wants the biogas generator to produce at least 3 000 kWh of electricity each week.
He collects 60 kg of dung from each cow per week.
1 kg of dung produces 0.095 kWh of electricity.

(i) Calculate how much electricity can be produced from each cow per week. [1]

Electricity produced = ... kWh

(ii) Calculate how many cows the farmer will need to produce the 3 000 kWh required. [1]

Number of cows = ...

(c) The table below shows the heating effect different greenhouse gases have on the atmosphere by comparing the global warming potential (GWP) values. If the GWP is twice as big the gas will cause twice the heating effect.

Greenhouse gas	GWP
carbon dioxide	1
methane	25
nitrous oxide	298

If it is left outside, cow dung releases **methane** into the atmosphere. In a biogas generator the methane is captured and burned releasing a similar amount of **carbon dioxide** (CO_2) into the atmosphere instead.

Alun suggests that biogas generators are bad for the environment because they release CO_2 into the atmosphere.
Explain whether you agree. [2]

....

....

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